ARC

Brewing Tea Device

Empathize

Role of AI: Clarifier

Ritual Interface ····· **Cultural Fusion** ····· **Material Warmth** ····· **Multiple brewing Methods**

"Beyond the technical specifications, ask me 10 critical questions that explore the hot beverage preparation rituals and sensory needs of users living in small spaces."

The AI defined the device not just as a "water heater," but as a "living space object." It focused on space constraints, safety concerns, and the "waiting psychology" involved in the preparation process.

App Integration: I rejected the mobile app idea because the ritual should be an analog break from screens, not something that brings the user back to their phone.

Stationary Focus: I accepted that the device should remain stationary on the desk like a "piece of furniture," rather than being picked up each time.

The "Wait Time": I transformed AI's fast boiling suggestion. I focused on a mechanism that makes the waiting time more enjoyable through a visual ritual.

Define

Role of AI:Reframer

"Reframe my user profile and relate the 6 reasons why people enjoy tea rituals to design features."

The AI reframed the user from a simple “water heater owner” to a “micro-ritualist,” shifting the design focus from a technical device to an emotional spatial anchor.

Design Decisions

Visual Weight: A minimalist silhouette and furniture aesthetic will be adopted to prevent the device from creating "visual clutter" in a confined space.

The Ritual Clock: The boiling time will be configured as a timer (clock) that determines the user's break time.

Tactile Safety: The exterior surface will be safe and textured to allow for the "hand-warming" ritual.

Ideate

Role of AI:Expander/Reframer

Focus: Resolving Tensions Through "Physics of Brewing"

Explain the real problems encountered while preparing tea and the different brewing techniques (Gongfu Cha, Decoction, etc.)."

Technical Issues: Acoustic noise during boiling, scalding (burning the tea at the wrong temperature), and dripping during pouring (the teapot effect).

Brewing Techniques: Each technique, from Western style (large volume) to Gongfu Cha (delicate and multiple brewing), demonstrated that it demands a different "geometry" and "heat control" from the device.

Cold Brewing Vibration:
I rejected the AI's suggested "vibration-assisted cold brewing" feature because it would disrupt the device's quiet and meditative nature.

Precision Temperature Control:
I acknowledge that precise temperature control is essential to preserve the delicate flavor of different tea types.

Prototype

Role of AI:Critique & Expander

Focus: The Integration of Precision & Ritual

Smoked Glass Tray:
I accepted the smoked glass surface because it makes cleaning easier and elegantly conceals limescale or stains.

The Interaction Hub:
I transformed the base from a simple electrical unit into a “Ritual Control Center,” where the scale, timer, and temperature control are managed.

Design Decisions

Visual Feedback: The integrated digital display under the glass shows the leaf quantity with 0.0g accuracy. This supports the "alchemical" aspect of tea preparation (such as the Gongfu Cha technique).

Rotary Knob and hybrid touch: The physical knob next to the display reduces digital clutter; it gives the user a haptic tactile sensation, allowing for temperature and time adjustment with the enjoyment of analog settings.

Test

Role of AI:Critique

Smoked Glass for Body:
In my form experiments in the PDF, I considered using a glass body. However, I rejected this idea because matte black stainless steel created a more dignified feeling of serenity and ritual. I only kept the glass on the tray.

Angled Handle:
Based on the ergonomic analysis in the Angled Handle PDF, I finalized the angled handle design to reduce pressure on the wrist.

Transformed the “Orientalist” Form:
To address the concern in the PDF about the form looking too orientalist (like a teacup), I widened the form downward and transformed it into a micro-architectural structure integrated with the lower station.

Design Decisions

By basing the design on a mass that expands downwards rather than upwards, the AI physically sealed the concept of 'Spiritual Spaciousness' that it suggests for the 'Micro-Ritualist' user.

Visual Process

Design Decisions

This tea brewer combines different tea and herb brewing methods in one device, allowing users to adjust herb portions and water temperature to shape the tea’s aroma and character. Blending modern technology with traditional tea rituals, it creates a calming moment after a busy day.
With precise dosing and smooth laminar flow brewing, the Arc pairs matte black steel with warm wood to create both tea and a sense of tranquility. Its display offers a hybrid interface that works as both a touchscreen and a rotary knob.

Design Decisions

*The tea brewing device completes its operation and indicates this with a yellow light as visual feedback. Additionally, a user interface will display a message on the screen indicating when the yellow light has run out.

Screen Interface

This tea brewing device offers different brewing styles on its interface and allows users to select the water and tea temperature. It also features an interface design that shows portioning, as the portioning of the herb used affects the aroma and taste of the tea. The interface includes screens displaying different brewing methods, providing the user with information about the different brewing types.